

Basic Principles of Medicine 1

Module : Foundation to Medicine

FTM DLA

DLA Title: Genomics in Diagnosis

Robert D. Finn PhD (rfinn@sgu.edu)

Acknowledgements: Mary Maj, PhD, Sharmila Upadhyaya, MBBS, MD, Andrew K.

Sobering, PhD

St Georges University ©

Learning Objectives:

SOM.MKIII.BPM1.1.FTM.3.GNET.0069	Explain how triplet repeat expansion disorders are diagnosed
SOM.MKIII.BPM1.1.FTM.3.GNET.0070	Discuss why each triplet repeat expansion disorder needs its own unique test for diagnosis
SOM.MKIII.BPM1.1.FTM.3.GNET.0071	Explain how a G-band karyotype is prepared
SOM.MKIII.BPM1.1.FTM.3.GNET.0072	Describe FISH and how it can be used in diagnostic applications
SOM.MKIII.BPM1.1.FTM.3.GNET.0073	Compare and contrast : Metaphase FISH, Interphase FISH, and Chromosome painting (SKY FISH, or spectral karyotype)
SOM.MKIII.BPM1.1.FTM.3.GNET.0074	Describe the strategy behind the microarray: CGH Microarray, SNP Microarray, cDNA Microarray

Detection of Triplet Repeat Expansion Disorders:

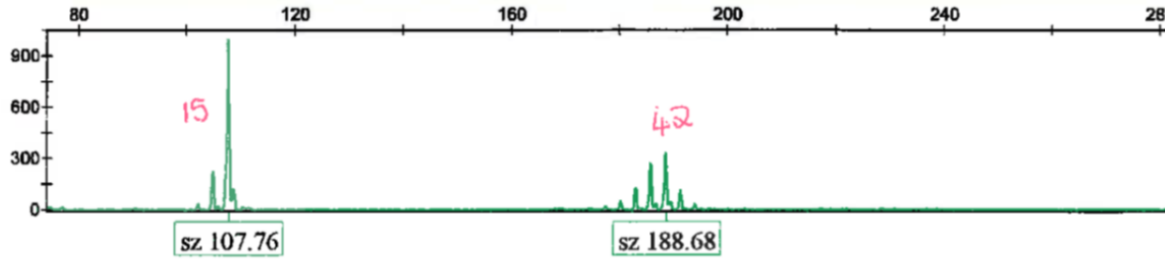
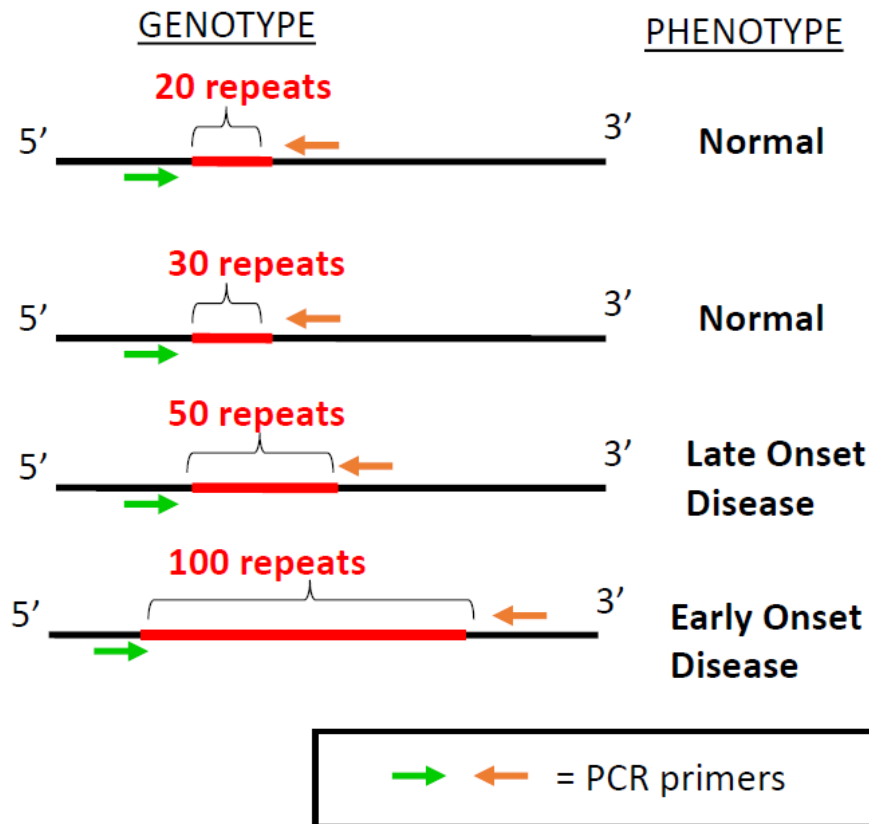
- Triplet repeat expansion disorders arise from amplification of a specific trinucleotide sequence in a specific location – Huntington disease, Fragile X syndrome, Myotonic dystrophy.
- The size of the repeat region is itself the diagnostic marker – more repeats = larger region!
- Each triplet repeat disorder requires its own diagnostic test.
- All tests based on a PCR reaction with specific primers flanking repeat region.
- Detect fragments by:
 1. Gel electrophoresis
 2. Column chromatography
 3. Southern blot – cut either side and detect using labelled probe
- Allows estimate of the number of repeat and comparison to threshold.

Detection of Triplet Repeat Expansion Disorders: contd

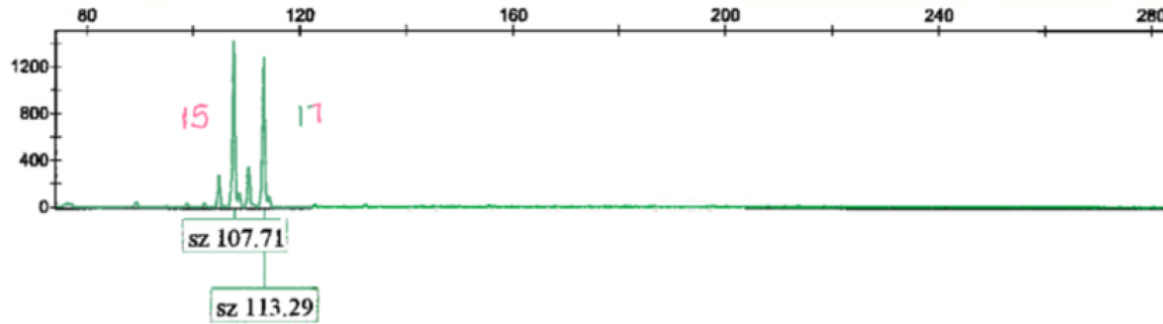
... CAG CAG CAG CAG CAG ...



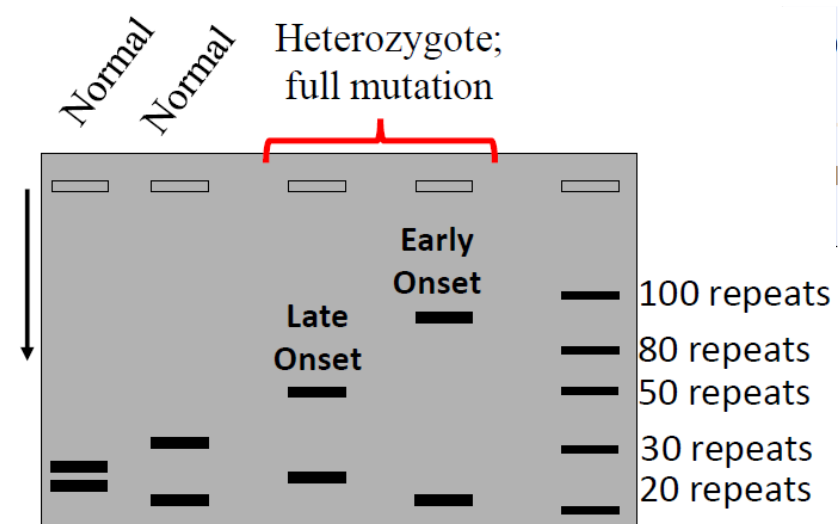
Huntington Disease gene contains a CAG repeat in the coding region.



Patient with Huntington disease

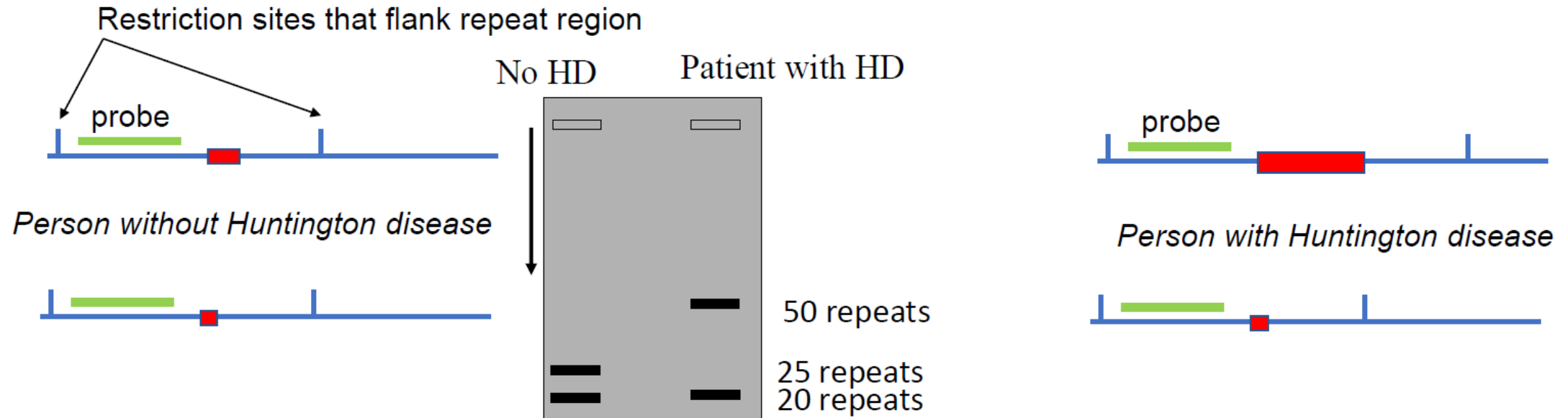


Patient without Huntington disease



Detection of Triplet Repeat Expansion Disorders: contd

Southern blot may diagnose triplet repeat expansion disorders

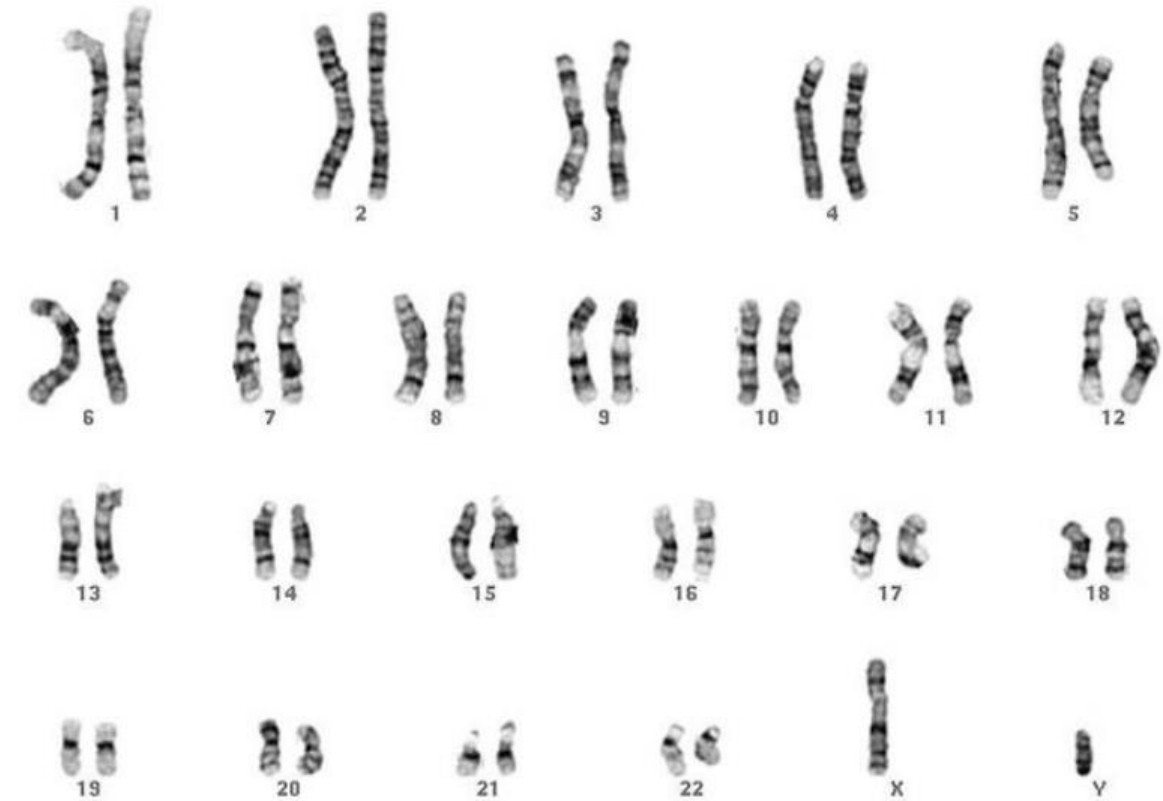


- Triple repeat disorders cannot be diagnosed using ASO, NGS or microarray.

G-Band Karyotyping:

- G-band karyotyping = picture of all chromosomes
- Detect – whole chromosome changes, very large deletions, duplication or rearrangements e.g. Down syndrome, Turner syndrome.
- Changes > 5 Mb, low resolution, depends on chromosomal location and change in pattern i.e. changes must be visible under light microscope (microdeletions/submicroscopic deletions).

Human male
G-bands



Preparation of G-Band Karyotype:

1. Isolate cells from peripheral blood
2. Culture cell in presence of phytohemagglutinin – stimulates cell division.
3. Arrest cells in metaphase using colchicine or colcemid.
4. Cells are swelled using a hypotonic solution – burst onto microscope slide.
5. Chromosomes fixed and prepared for staining and microscope visualization.
6. Giemsa staining – regions of dark and light (characteristic pattern).

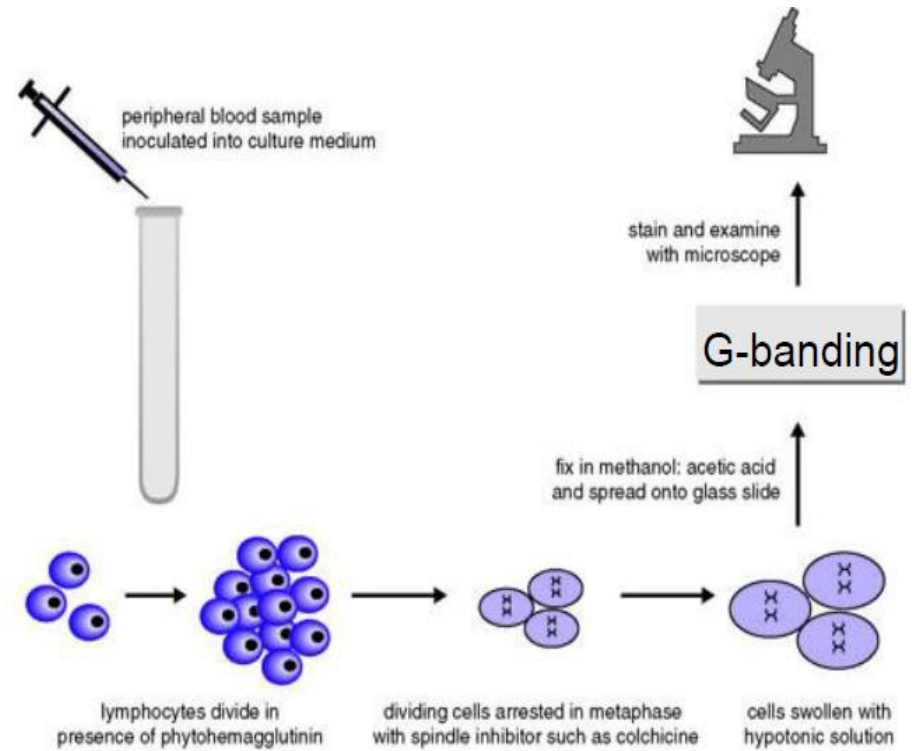
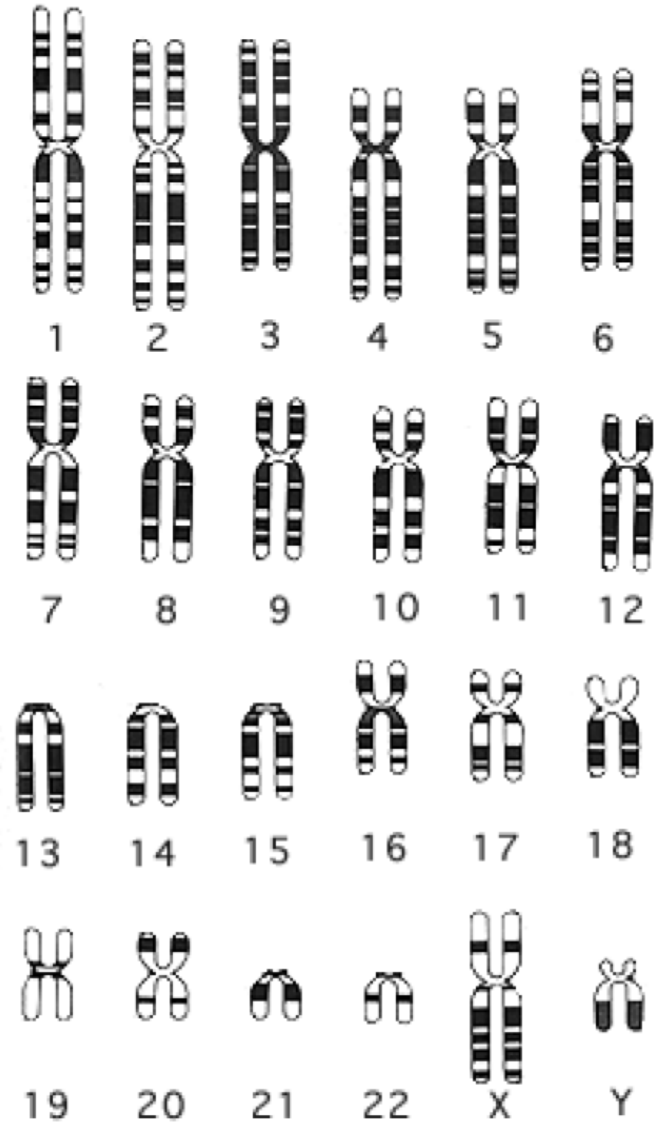


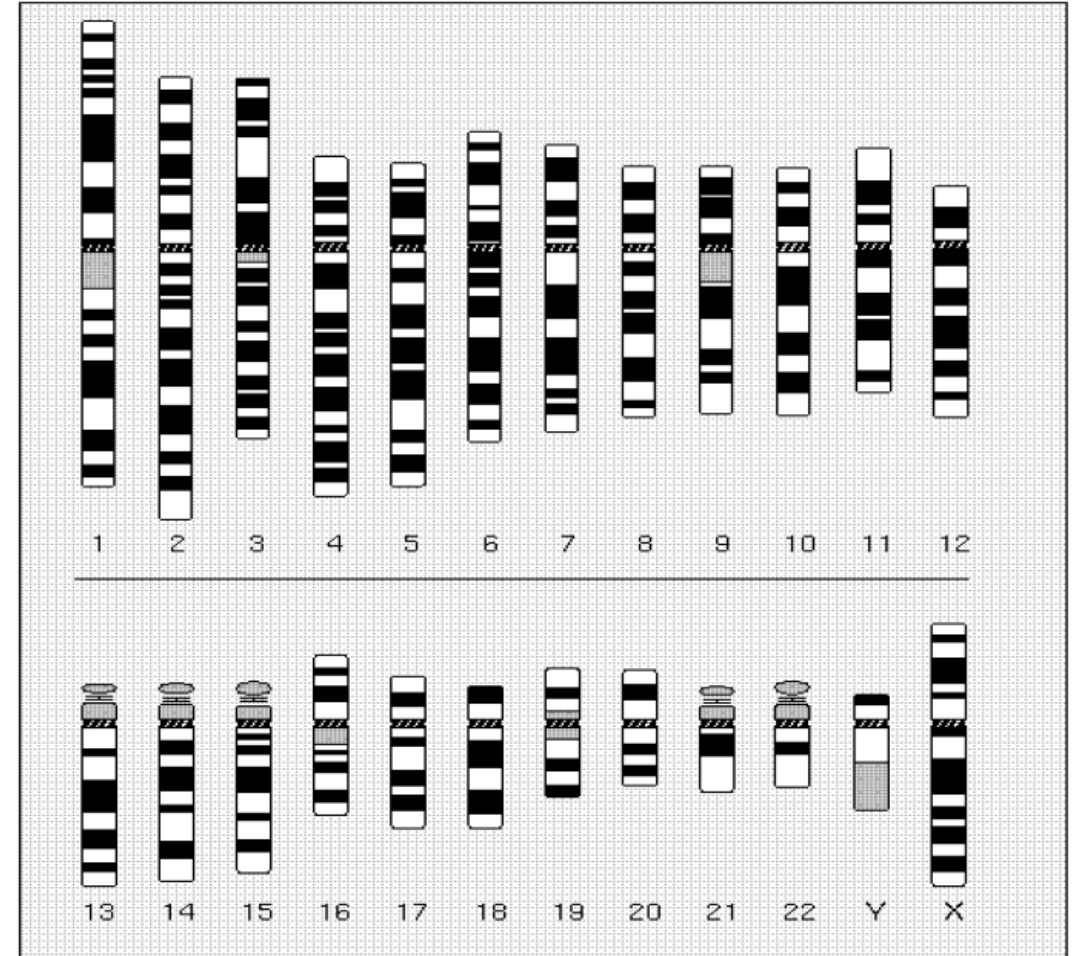
Figure from Korf and Irons – chapter 6



Ideogram:

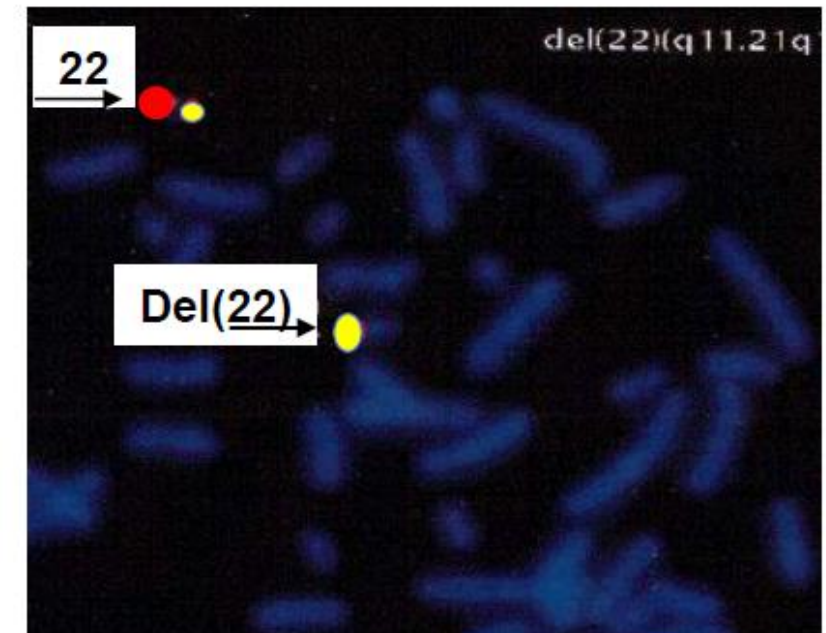
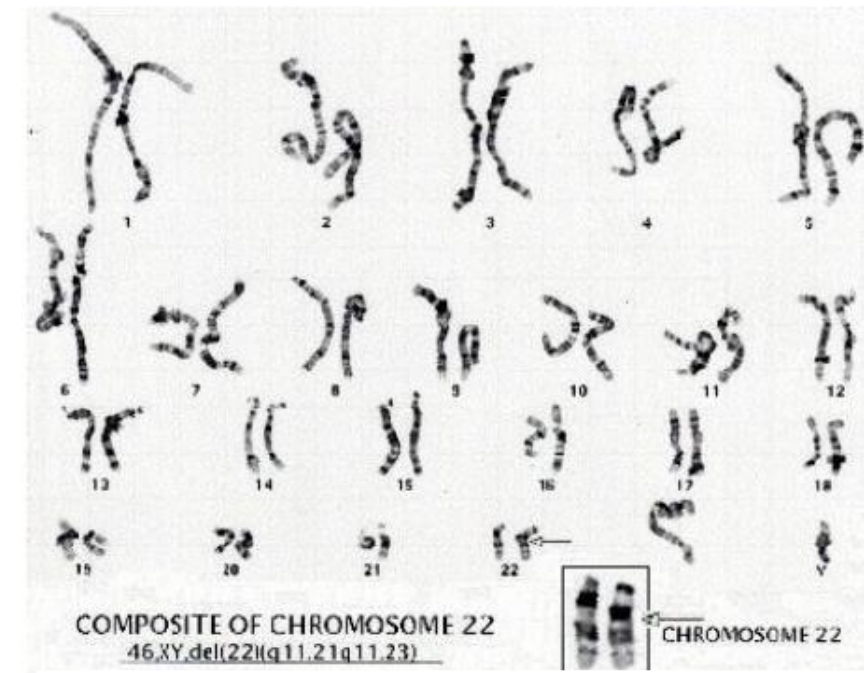


Schematic
diagram
(**ideogram**) of
the set of
normal human
metaphase-stage
chromosomes



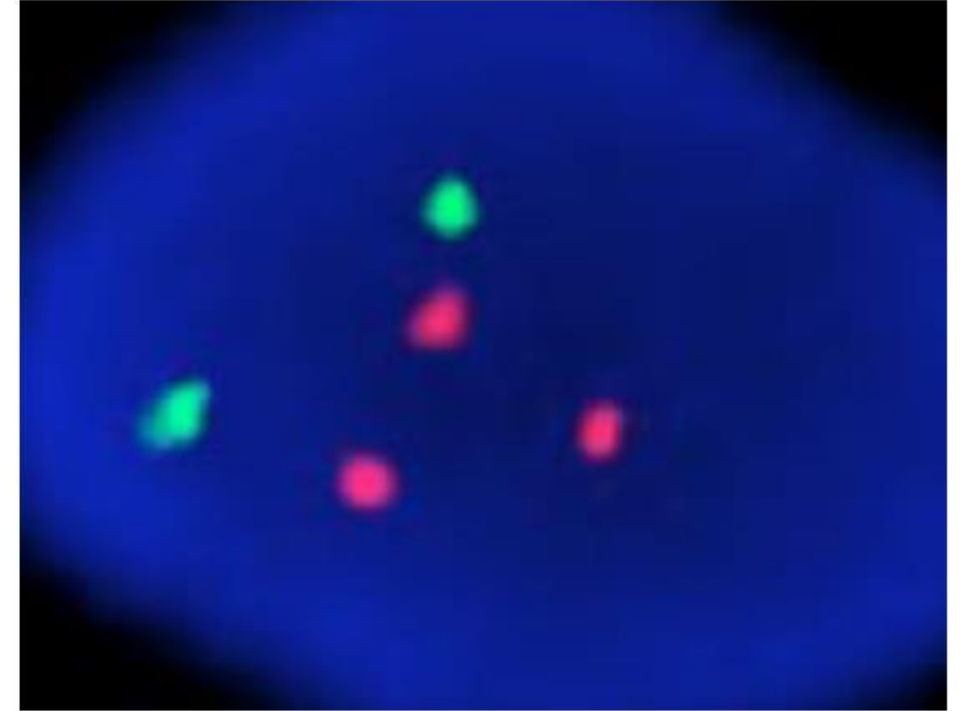
Fluorescent In Situ Hybridization (FISH):

- Metaphase chromosome spread prepared as for G-Band Karyotyping – “stain” with specific single strand probe (100 – 500 kb), not Giemsa.
- Two types:
 1. **Chromosome/Gene specific** – probe designed to a specific locus (hypothesis driven), detect submicroscopic deletions, translocations and duplications.
- Resolution higher than G-band Karyotyping e.g. DiGeorge/22q11.2 deletion syndrome – only occasionally diagnosable via G-banding, easier to detect with FISH using probe for deleted region.



Interphase FISH:

- Interphase stage cells can be used for certain conditions – aneuploid (13, 18 or 21) and sex determination of a baby (X and Y).
- Term Rapid FISH as cell culture is not required (no metaphase arrest) – very fast turnaround time.
- Used two different probes to two different chromosomes e.g. chr 13 (**green**) and chr 21 (**red**) – patient is trisomy 21 i.e. Down syndrome.



Chromosome painting/SKY (Spectral Karyotyping):

- Two types:

2. Chromosome painting/SKY (Spectral Karyotyping) – whole chromosome specific probes (different probe/color per chromosome), entire genome analysis for complex rearrangements.

- Probes unique to each chromosome are created by labelling with different colored fluorescent molecules.
- Different probes mixed and hybridized to metaphase stage chromosome spreads.
- Labelled chromosomes analyzed by fluorescent microscopy.

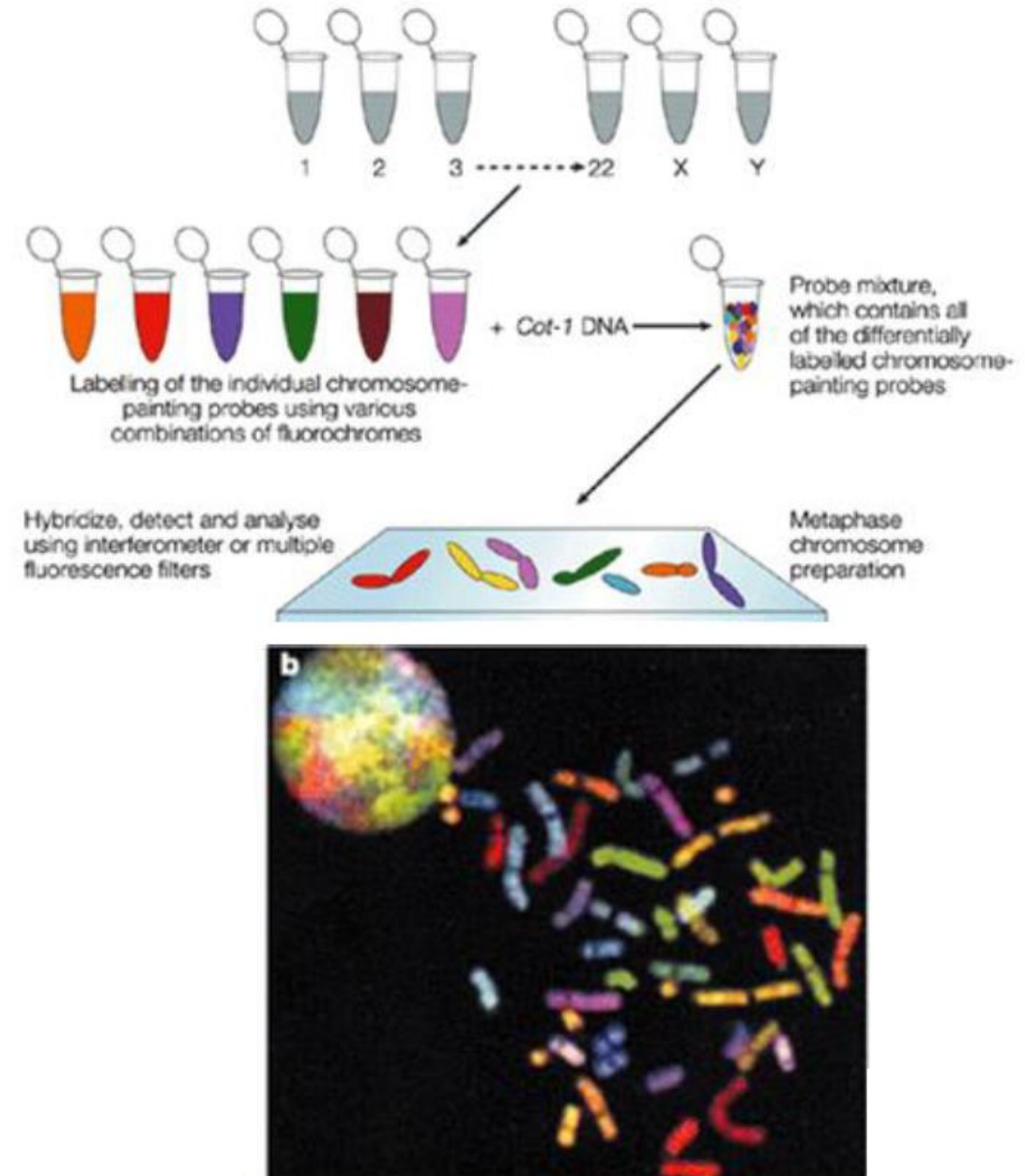
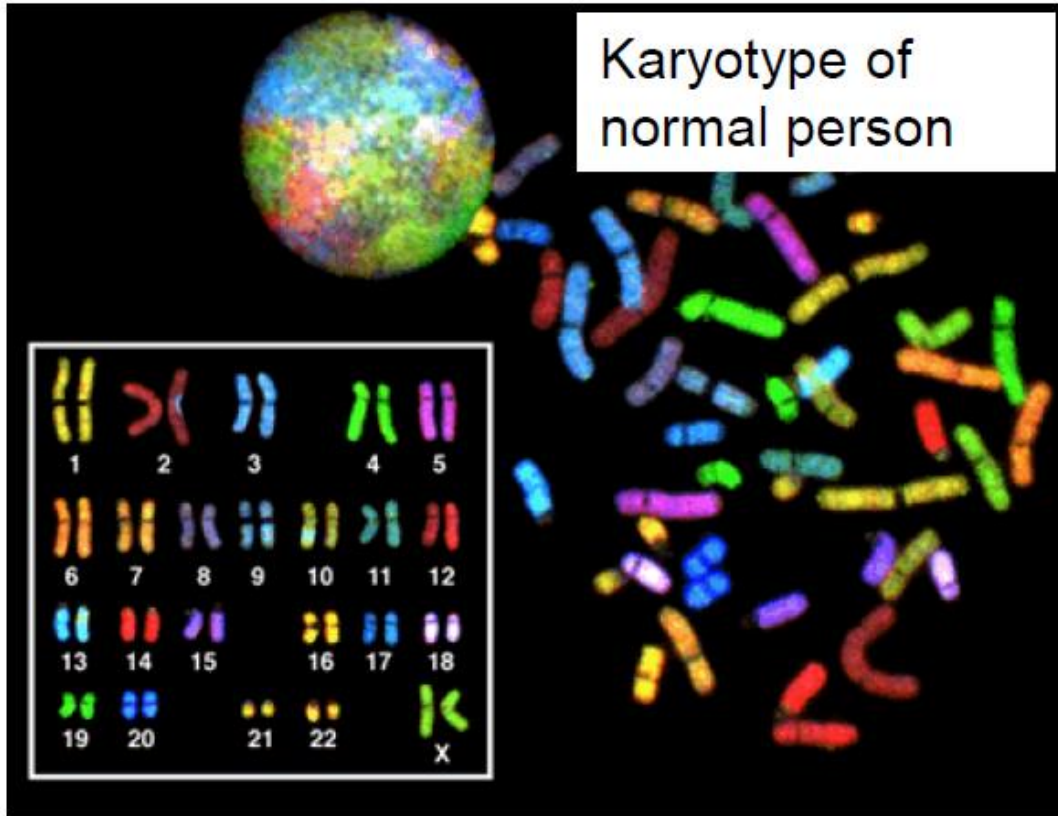


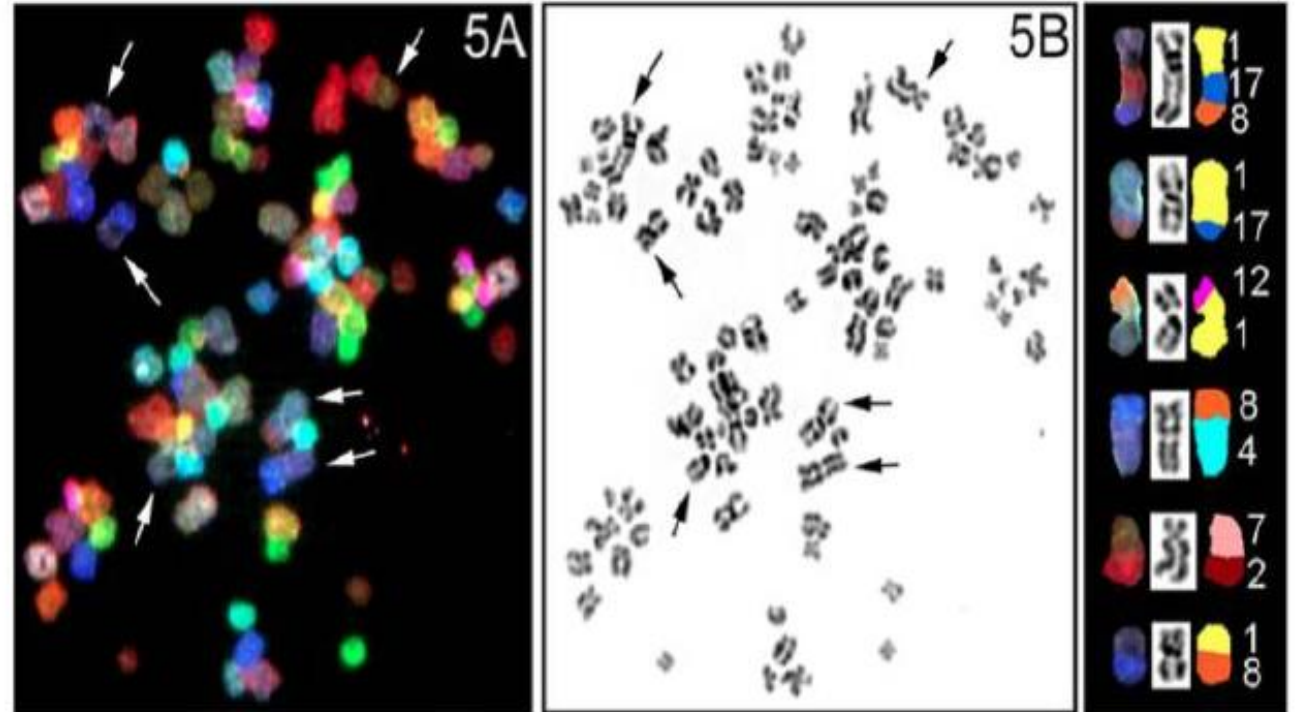
Image attribution: National Human Genome Research Institute; free use

SOM.MKIII.BPM1.1.FTM.3.GNET.0072/0073

Chromosome painting/SKY (Spectral Karyotyping): contd



- Deletion or duplications not visible due to single color per chromosome.
- Use in cancer genetics to visualize genomic instability.

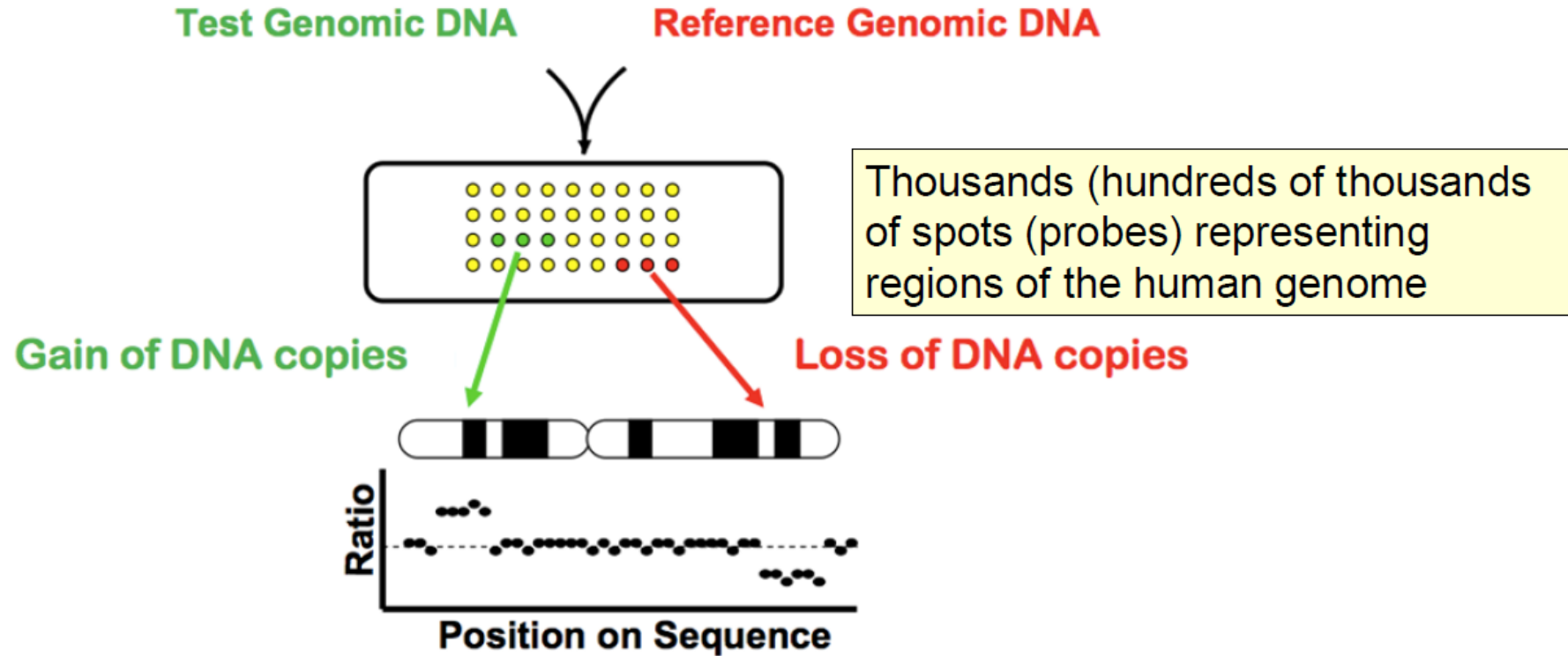


Microarrays:

- Comparative Genome Hybridization (CGH) - detects deletion and duplications (lower limit 30 kbps), greater resolution than G-Banding (not balance rearrangements, Triple repeats, SSR or SNPs).
- Single Nucleotide Polymorphism (SNP) – detects single nucleotide difference = individual genetic fingerprint/profile.
- Expression or cDNA – detect gene expression in a specific tissue/cell at a specific time/under specific conditions.
- All based on principle of hybridization between sample DNA and a specific probe – one is labelled.

Comparative Genome Hybridization Microarray:

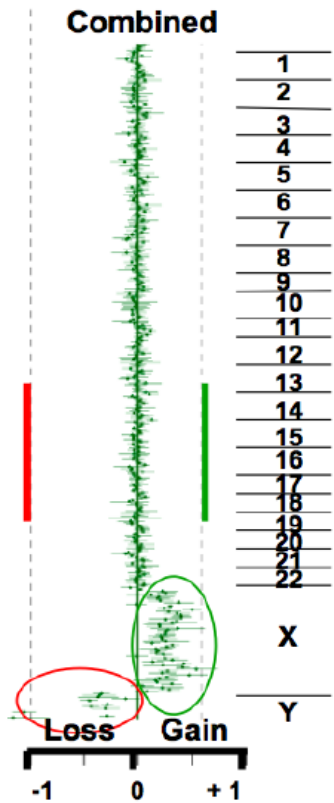
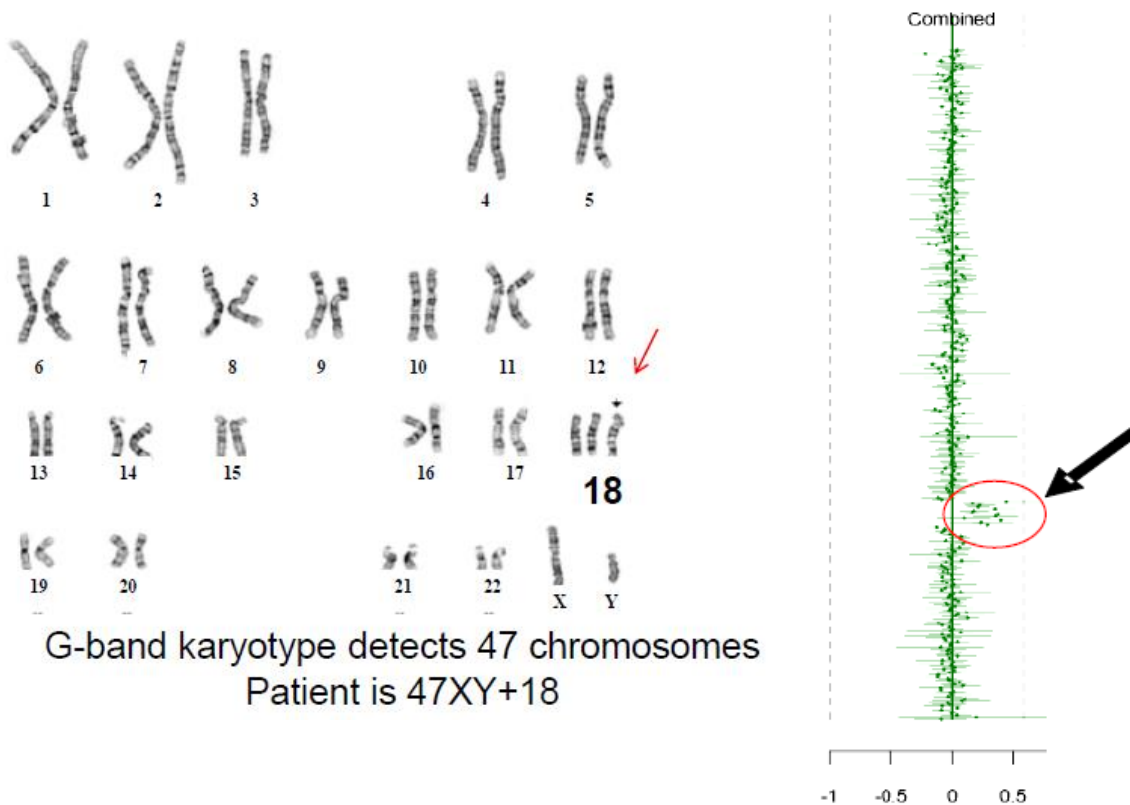
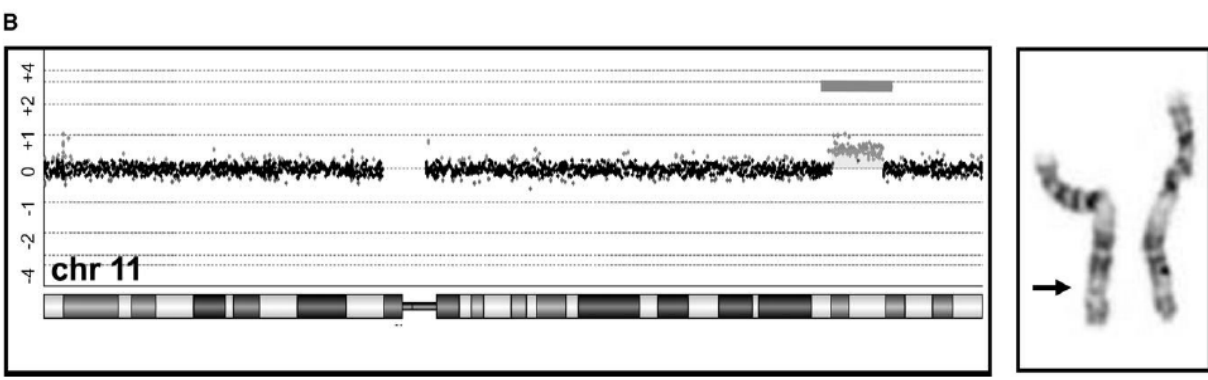
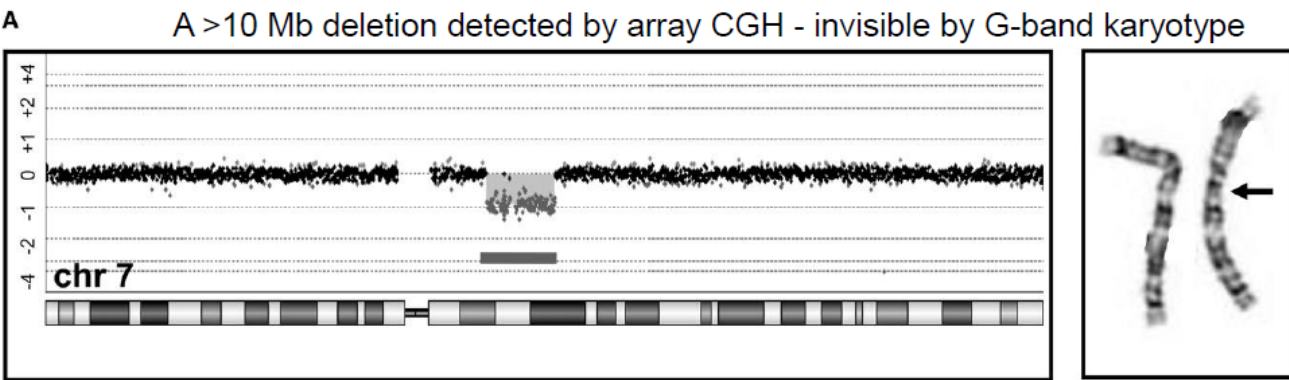
- CGH – based on co-hybridization of two differently labelled DNAs to a microarray.
- One sample = reference, other = test/patient – allows comparison to reference e.g. tumor to normal.



- Probes spread across the entire genome – number of probes represents resolution.



CGH Examples:

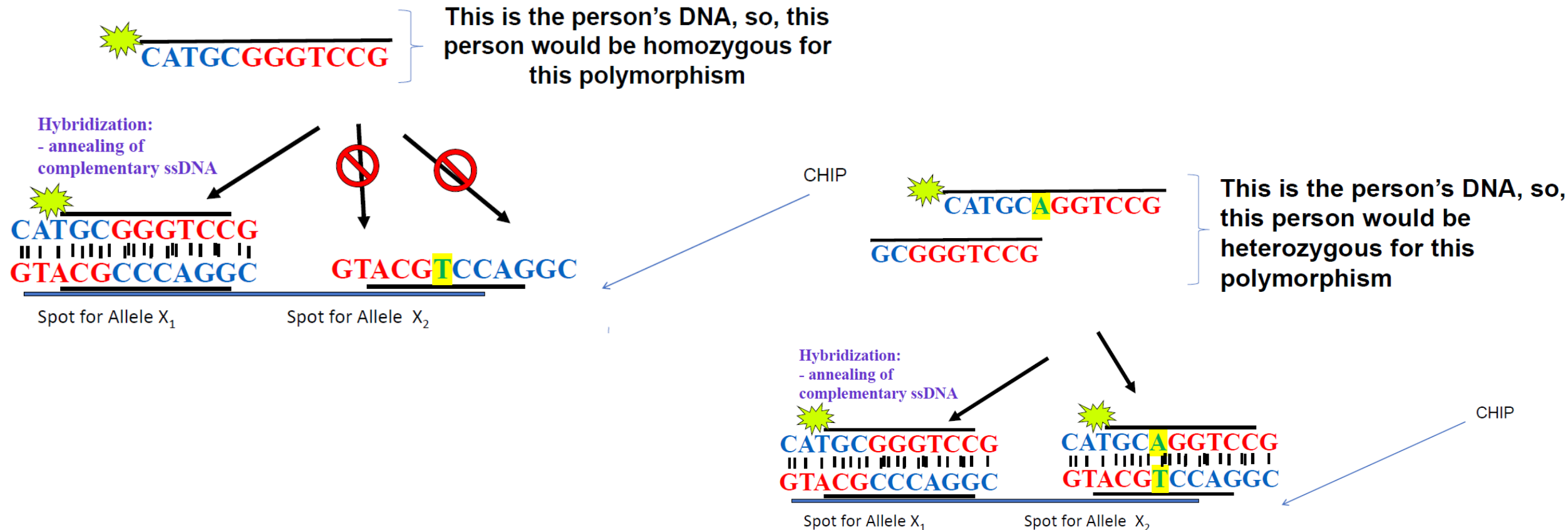


Comparison of
normal male
and normal
female.

G-band karyotype detects 47 chromosomes
Patient is 47XY+18

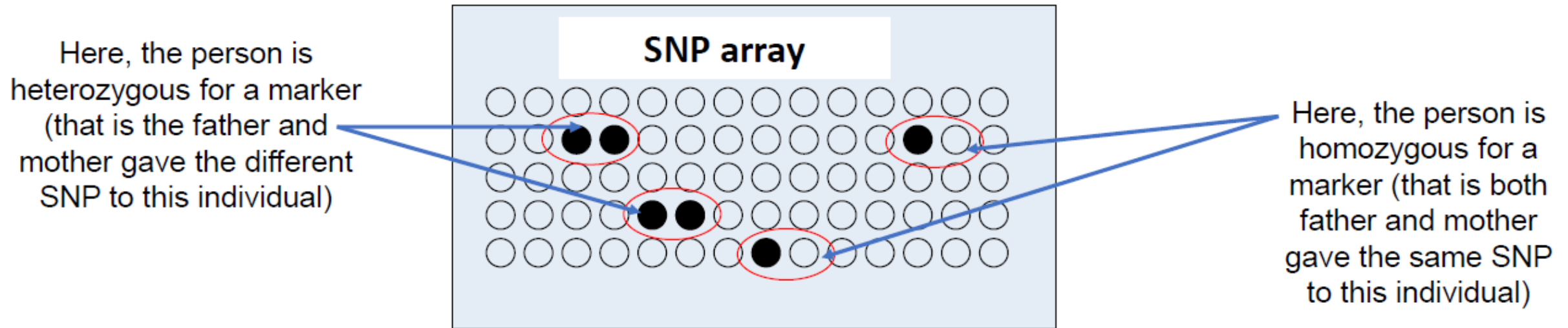
Single Nucleotide Polymorphism Microarray:

- SNP – single nucleotide variations, 1 every 1000bp in human genome = very common.
- Isolate genomic DNA, make single stranded, fragment and label with fluorescent tag.
- Hybridize labelled genomic DNA to immobilized probes on microarray single – each probe = various alleles of a corresponding gene.



SNP Microarray contd:

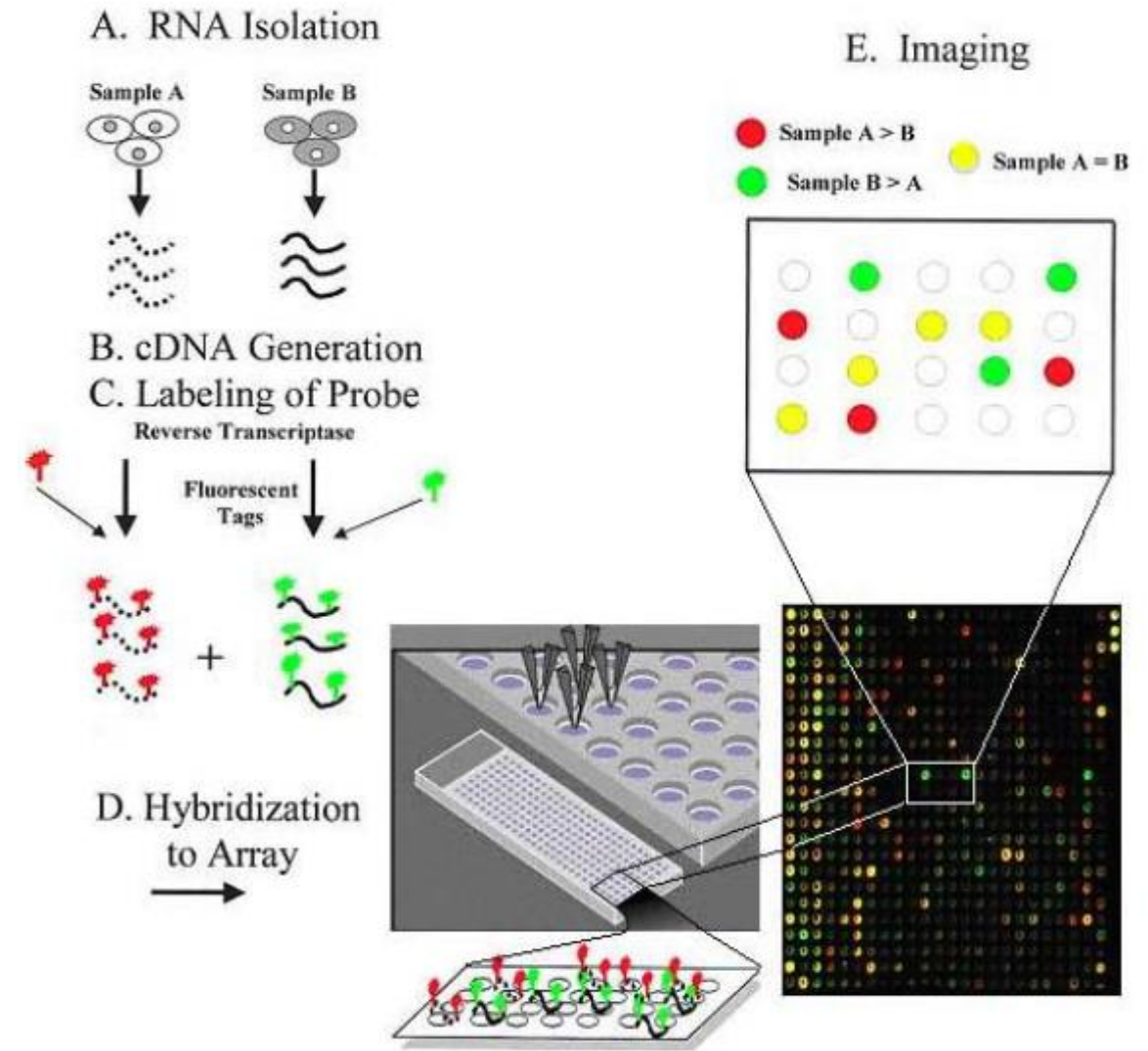
- Allele probe arranged in pairs – allows determination of homo/heterozygosity.
- Positive signal = allele present, blank = absent.
- Microarray contains thousands/millions of probes – genetic profile in single experiment.



- Key application is in GWAS (Genome Wide Association Studies) – potential correlation between SNP and disease.

cDNA/Expression Microarray:

- Analysis of genes transcribed in a particular situation.
- **Requires mRNA to be converted to cDNA** – cDNA labelled with fluorescent tag.
- Two potential goals:
 1. Gene **expression** in particular cell or tissue.
 2. Comparative – expression difference between different cell/tissue samples or same cell/tissue under different conditions e.g. time, drug treatment.
- In comparative, color relates to relative abundance!



Summary:

- Triplet repeat disorders:
 - Arise from the amplification of a specific trinucleotide sequence at a specific location.
 - Detect by PCR and fragment size comparison.
- G-Band Karyotyping:
 - Equates to a picture of chromosomes obtained from mitotic arrested cells.
 - Detect increase chr number, large deletion/duplication and rearrangements (must be detectable by light microscope).
- FISH:
 - Specific fluorescent probe for specific locus or chr (metaphase or interphase).
 - Detect submicroscopic deletions, duplications, rearrangements
- Microarrays:
 - CGH (chr deletions/duplications), SNP (single nucleotide variations – fingerprint), Expression (gene transcription – specific location/conditions).

Thank you for listening!